

KAZI EHSANUL KARIM

Molecular Simulation | Nanoscale Transport | Multiscale Thermo-Fluids | Machine-Learning Interatomic Potentials

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Postdoctoral / Research Fellow Opportunities | Target start: early 2027

RESEARCH PROFILE

Computational researcher and Ph.D. candidate in Mechanical Engineering at the University of Ulsan, with three first-author articles in Physical Review E and Physical Chemistry Chemical Physics (two published, one accepted). Combines molecular dynamics, statistical mechanics and continuum modelling to investigate mass and energy transport under nanoscale confinement. Current independent work builds and tests machine-learned interatomic potentials for CNT-confined NaCl–water, with a focus on diagnosing model error. Research interests include error-guided active learning, physics-informed modelling and agentic simulation workflows.

Research methods: LAMMPS EMD/NEMD; CP2K; DP-GEN / DeePMD-kit; Green–Kubo transport; continuum boundary conditions; Python / Linux / HPC; ANSYS thermo-fluids.

EDUCATION

University of Ulsan, Republic of Korea Mar 2019 – Present

Ph.D. Candidate, Integrated M.S.–Ph.D. in Mechanical Engineering | Micro/Nano-scale Thermo-Fluids Engineering

Laboratory | Supervisor: Prof. BoHung Kim

Dissertation defense planned: November 2026 | Formal degree conferral expected: February 2027

Dissertation: What Survives Sub-nm Confinement? — The Meaning of Slip Length under Molecular Accessibility, Ergodicity, and Structure–Transport Decoupling.

Khulna University of Engineering & Technology (KUET), Bangladesh

May 2015

B.Sc., Mechanical Engineering

DOCTORAL RESEARCH EXPERIENCE

Doctoral Researcher — Molecular Simulation & Nanofluidics, University of Ulsan Mar 2019 – Present

- **Boundary thermodynamics** — Developed a first-law energy-balance analysis of the nanoscale flow boundary and tested NEMD against Hagen–Poiseuille limits under controlled wall–fluid interaction (Physical Review E, 2021).
- **Water transport in CNTs** — Critically assessed scale effects and boundary definitions in CNT water-transport models, relating molecular-simulation evidence to continuum predictions (Physical Chemistry Chemical Physics, 2024).
- **Confinement and transport** — Analysed molecular admission, packing and transport in subnanometre graphene slits (Physical Chemistry Chemical Physics, accepted 2026); investigate finite-time diffusivity, hydraulic resistance and continuum-closure limits in carbon nanotubes and graphene nanoslits.
- **Reproducibility and uncertainty** — Apply ensemble, timestep, box-size, forcing and sampling checks; block averaging and confidence intervals; and thermostat-sensitivity and cross-method comparisons to test transport interpretations.

INDEPENDENT RESEARCH PROJECT

Error-Aware MLIPs for CNT-Confined NaCl–Water 2026 – Present

- **Workflow** — Built and ran an end-to-end workflow for NaCl–water in four carbon-nanotube diameters — CP2K (PBE-D3(BJ)) ab initio MD seeds and reference labels → DP-GEN → DeePMD-kit → LAMMPS — with a verification step at every stage.
- **Evaluation** — Developed species-resolved force validation on fresh-DFT holdout frames; identified a species-specific failure mode that the aggregate error concealed and committee disagreement did not flag; tested targeted reference-data augmentation (error unchanged; cause under investigation).
- **Transport** — Computed equilibrium (Green–Kubo) interfacial friction from MLIP-driven MD with block-bootstrap uncertainty and model-disagreement screening of trajectories; two standard estimators disagree, and the cause is under investigation.
- **Tooling** — Wrote a tested Python tool-kit for per-species force metrics, group-aware uncertainty, model-scope checks and rules-based decisions; public methods demonstration (synthetic data) at [kazi21.github.io](#).

PUBLICATIONS & MANUSCRIPTS

Peer-reviewed journal articles — published and accepted

1. K. E. Karim, B. Kim, "Molecular admission and packing govern transport in subnanometre graphene slits," *Physical Chemistry Chemical Physics*, accepted for publication (2026).
2. K. E. Karim, M. Barisik, C. Bakli, B. Kim, "Estimating water transport in carbon nanotubes: A critical review and inclusion of scale effects," *Physical Chemistry Chemical Physics*, 2024, 26(28), 19069–19082. [DOI](#)
3. K. E. Karim, B. Kim, "First law of thermodynamics on the boundary for flow through a carbon nanotube," *Physical Review E*, 2021, 103(5), 053115. [DOI](#)

Manuscripts — completed and in final preparation

4. K. E. Karim, B. Kim, "Wall–fluid interaction strength shifts the slit width of maximum finite-time diffusivity in a graphene-confined simple fluid," manuscript completed; preparing submission.
5. K. E. Karim, B. Kim, "Confinement-Dependent Hydraulic Resistance and Limits of Navier-Slip Transferability in Graphene Nanoslits," manuscript in final preparation.

In preparation — current research directions

Physics-informed neural networks (PINNs) for structure–friction–flow closure; multimodal electrolyte transport in carbon nanotubes; and error-aware machine-learning interatomic potentials (MLIPs) for agentic multiscale simulation.

Additional peer-reviewed work

- A. S. M. M. Rahman, K. E. Karim, M. H. S. Simanto, "Effect of heat treatment on low carbon steel: An experimental investigation," *Applied Mechanics and Materials*, 2016, 860, 7–12. [DOI](#)

Also: one additional journal article on RPR manipulator modelling and five peer-reviewed conference papers in thermal, materials, and design engineering.

TECHNICAL EXPERTISE

- **Molecular simulation** — LAMMPS; EMD/NEMD; NVT/NPT/NVE; confined fluids; pressure-driven and Couette flow; Green–Kubo and NEMD transport coefficients; interfacial friction/slip; spatial binning and profile analysis.
- **Electronic structure & MLIP** — CP2K periodic GPW reference calculations using PBE-D3(BJ), GTH pseudopotentials and DZVP-MOLOPT basis sets; DP-GEN active learning; DeePMD-kit dataset curation, committee training, holdout validation, error analysis, stability testing and LAMMPS deployment.
- **Scientific computing & statistics** — Python (NumPy, SciPy, pandas, Matplotlib, MDAnalysis, scikit-learn), Bash, Linux/HPC, MPI workflows, automated parameter sweeps; block averaging, confidence intervals, convergence/sensitivity diagnostics, uncertainty-aware regression.
- **CFD & continuum modelling** — ANSYS geometry/mesh generation, compressible nozzle flow, boundary-condition specification, convergence monitoring, and pressure/density/temperature/velocity/Mach post-processing; nozzle-wall and coolant-passage thermal analysis.

ADDITIONAL PROJECTS & EXPERIENCE

Thermo-Fluid Simulation of a Methane Rocket-Engine Nozzle & Cooling Passage

2022

Industry-linked laboratory project using ANSYS for geometry/mesh generation, compressible nozzle-flow and high-temperature thermal analysis, and post-processing of density, pressure, temperature, velocity, and Mach-number fields; included nozzle-wall and methane coolant-passage thermal behaviour.

Research Assistant — Robotics & Mechatronics Laboratory, KUET

Feb 2014 – May 2015

Kinematic modelling of an RPR manipulator, automated-wheelchair design, and experimental thermal/materials studies.

TEACHING & MENTORING

Practical MLIP Crash Course (CP2K → DeePMD-kit → LAMMPS) — course developer

2026 – Present

- **Curriculum** — Developing a 30-lesson practical course for senior undergraduates and first-year M.S. students: runnable recipes, clearly labelled synthetic examples, a break-it-and-fix-it case in every lesson (worked solutions complete for six core lessons) and an interview playbook; learner pilot planned.

REFERENCES

References available upon request.